

BCSE318L	DATA PRIVACY			L	T	P	C
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Pre-requisite	NIL	Syllabus version					
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Course Objectives							
1. To impart the need of data privacy.							
2. To categorize the statistical and computational techniques required to share data, with a primary focus on the social, and health sciences.							
3. To formulate architectural, algorithmic, and technological foundations for the maintaining the data privacy.							
Course Outcomes							
After completion of this course, the student shall be able to:							
1. Characterize basic rules, principles for protecting privacy and personally identifiable information.							
2. Formulate data that supports useful statistical inference while minimizing the disclosure of sensitive information.							
3. Identify the list of threats on the various types of anonymized data.							
4. Classify and analyze the methods of test data generation with Privacy and utility.							
Module:1	Data privacy and Importance					5 hours	
Need for Sharing Data - Methods of Protecting Data - Importance of Balancing Data Privacy and Utility – Disclosure - Tabular Data - Micro data - Approaches to Statistical disclosure control – Ethics – principles - guidelines and regulations.							
Module:2	Microdata					7 hours	
Disclosure - Disclosure risk - Estimating re-identification risk - Non-Perturbative Micro data masking - Perturbative Micro data masking - Information loss in Micro data.							
Module:3	Static Data Anonymization on Multidimensional Data					7 hours	
Privacy – Preserving Methods - Classification of Data in a Multidimensional Dataset - Group-based Anonymization: k-Anonymity, l-Diversity, t-Closeness.							
Module:4	Anonymization on Complex Data Structures					8 hours	
Privacy-Preserving Graph Data, Privacy-Preserving Time Series Data, Time Series Data Protection Methods, Privacy Preservation of Longitudinal Data, Privacy Preservation of Transaction Data.							
Module:5	Threats to Anonymized Data					6 hours	
Threats to Anonymized Data, Threats to Data Structures, Threats by Anonymization Techniques: Randomization, k-Anonymization, l-Diversity, t-Closeness.							
Module:6	Dynamic Data Protection					5 hours	
Dynamic Data Protection: Tokenization, Understanding Tokenization, Use Cases for Dynamic Data Protection, Benefits of Tokenization Compared to Other Methods, Components for Tokenization.							
Module:7	Privacy-Preserving Test Data Generation and Privacy Regulations					5 hours	
Test Data Fundamentals - Insufficiencies of Anonymized Test Data. Privacy regulations: UK Data Protection Act, Swiss Data Protection Act, HIPPA, General Data Protection Regulation.							
Module:8	Contemporary Issues					2 hours	
	Total Lecture hours:					45 hours	
Text Book							
1.	NatarajVenkataramanan, AshwinShriram, Data Privacy: Principles and Practice, 2016, 1st Edition, Taylor & Francis. (ISBN No.: 978-1-49-872104-2), United Kingdom.						